Unit Test Specification

Collaborative Study Material and Quiz Platform
Software Engineering Mini Project Team

Team Members

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Abstract—This document defines concise unit tests for the core Python functions of the Collaborative Study Material and Quiz Platform. Each test isolates one function or small service behavior.

I. Introduction

Unit testing verifies individual functions before they are combined into a complete application. The existing test suite uses Python unittest and a test database initialized before each test.

II. Unit Test Cases

ID	Unit Under Test	Input	Expected Result
UT-01	create_user	Valid username, password, Student role	User is created
UT-02	authenticate_user	Created user credentials	User record is returned
UT-03	create_question	Question, options, answer, topic	Question is stored
UT-04	submit_quiz_attempt	Correct answer for quiz	Score equals 100%
UT-05	create_test	Empty title or questions	ValueError is raised

III. Test Procedure

Initialize the database, create the required fixture data, call the target function, and compare the result with an assertion. Close database connections after each operation.

IV. Conclusion

These tests provide fast feedback for authentication, data creation, validation, and quiz evaluation. They should run before integration or system tests.